

Students' Stress Scores Prediction by Using ML

Hafsa Zulfiqar¹ Hafsa Ashraf² Kamran Saeed³

Department of Computer Engineering

Bahria University of Engineering and Applied Sciences

Abstract

Student mental health has become a growing concern in higher education institutions worldwide. Stress among university students adversely affects academic performance, physical health, and overall well-being. In this study, we conduct a comparative analysis of three machine learning regression models — Linear Regression (LR), Support Vector Regression (SVR) with Radial Basis Function (RBF) kernel, and Random Forest Regression (RF) — for predicting a continuous Stress Score from a dataset of 3,000 university students. The dataset comprises 16 features including demographic, academic, lifestyle, and psychological factors. Experiments were conducted in both Python (scikit-learn) and WEKA environments using an 80/20 train-test split. Our results show that Linear Regression achieved the highest performance with $R^2=0.85$, MAE=2.21, RMSE=2.54, and categorical accuracy of 81.33%, followed by SVR-RBF ($R^2=0.80$, Acc=79.50%) and Random Forest ($R^2=0.76$, Acc=79.83%). Screen Time, Exam_Frequency, Family Support, Assignment Load, and Social Media Use emerged as the most influential predictors. These results can help for university administrators and mental health practitioners seeking data-driven early-warning systems for at-risk students.

Keywords: Student Stress Prediction, Machine Learning, Linear Regression, Support Vector Regression, Random Forest, Mental Health,, Educational Data Mining

1 Introduction

Mental health disorders among university students represent a significant and escalating public health challenge. Epidemiological surveys estimate that between 30% and 40% of university students experience clinically significant levels of anxiety, depression, or chronic stress during their academic careers [1]. The consequences of unmanaged academic stress extend far beyond subjective discomfort: chronic stress is empirically

associated with cognitive impairment, reduced academic performance, immune suppression, sleep disturbances, substance use, and in severe cases, suicidal ideation [2,9].

The traditional approach to identifying stressed students relies on self-reported questionnaires administered at fixed intervals, counselor referrals triggered by academic failure, and reactive crisis interventions — all of which suffer from underreporting due to stigma, limited clinical resources, and inherent delays between stress onset and institutional response [3]. Machine learning (ML) offers a promising complementary paradigm: by analysing routinely observable features such as academic workload, sleep patterns, screen time, and family support, predictive models can flag at-risk students proactively, enabling earlier and more targeted interventions [4].

Regression-based ML approaches are particularly suited for predicting a continuous stress score rather than assigning students to broad Low/Medium/High categories, as they preserve the full granularity of stress measurement and enable clinical monitoring of stress trajectories over time [5]. This study introduces a complete, end-to-end ML-based student stress prediction system, encompassing data preprocessing, model training and evaluation, cross-platform validation, and deployment as an interactive Streamlit dashboard. Three representative regression paradigms are evaluated:

- Comparing three regression algorithms — Linear Regression, SVR (RBF kernel), and Random Forest — on a 3,000-record university student stress dataset.

- Evaluating models using five regression metrics: Correlation Coefficient (r), MAE, RMSE, RAE, and RRSE.

- Identifying the most influential predictors of student stress via Random Forest feature importance.

- Validating results across Python/scikit-learn and WEKA platforms to ensure reproducibility.

- Providing detailed visual analysis through scatter plots, line plots, bar charts, and distribution histograms.

A dataset of 3,000 university students with 15 features spanning academic, lifestyle, social, and demographic domains was used. Models were cross-validated using both Python scikit-learn and the Weka machine learning workbench. Results were visualised

Predicting Student Stress Using Regression Models and deployed via a professional dark-themed Streamlit dashboard featuring real-time prediction, gauge charts, scatter plots, confusion matrices, and feature importance visualisations.

The remainder of this paper is structured as follows: Section 2 reviews related work; Section 3 describes the dataset and preprocessing pipeline; Section 4 details the methodology; Section 5 presents experimental results; Section 6 provides comparative discussion; Section 7 describes the Streamlit dashboard; and Section 8 concludes with future directions.

2 Literature Review

2.1 Theoretical Foundations of Academic Stress

The scientific study of psychological stress has its roots in Hans Selye's General Adaptation Syndrome [7], which characterised stress as a non-specific physiological response to perceived demands. Lazarus and Folkman subsequently introduced the highly influential transactional model [6], positing that stress arises not from objective demands alone, but from the individual's cognitive appraisal of those demands relative to their available coping resources. This appraisal-based framework has profound implications for student stress research: two students facing identical examination schedules may experience markedly different stress levels depending on their perceived self-efficacy, social support, and prior academic experiences.

In the university context, Robotham and Julian [8] systematically reviewed the academic stress literature and identified five consistent stressor domains: academic workload and time pressure, examinations and assessments, financial concerns, career uncertainty, and social and interpersonal difficulties. Their meta-analysis found that academic stressors (workload, exams) were more consistently predictive of stress outcomes than environmental stressors, a finding largely corroborated by Pascoe et al. [9] in a more recent systematic review covering 2000–2018.

Critically for the present study, Thoits [5] demonstrated through a comprehensive review that social support — encompassing emotional, informational, and instrumental support from family and peers — acts as a significant moderator and mediator of the stressoutcome relationship, reducing both the appraisal of demands as threatening and

the

physiological stress response itself. This theoretical grounding directly predicts the protective role of Family_Support observed in our regression models.

2.2 Epidemiology of University Student Mental Health

The global burden of mental health problems among university students is welldocumented. The WHO World Mental Health International College Student (WMH-ICS) project [1], spanning 19 universities across eight countries, found that 35% of first-year students met diagnostic criteria for at least one DSM-IV disorder, with anxiety disorders (20.4%) and mood disorders (21.4%) most prevalent. Notably, the majority had not received any treatment, highlighting the critical gap between need and service provision.

In the United Kingdom context, Beiter et al. [2] administered validated instruments to 374 undergraduate students and found that 45.7% reported levels of stress severe enough to warrant clinical concern, with 40.9% reporting significant anxiety. Exam performance, academic workload, and post-graduation plans were the top three self-reported stressors. Eisenberg et al. [3] used regression analysis on a large multi-campus US sample to demonstrate that even sub-clinical levels of depression and anxiety were associated with GPA reductions of 0.1–0.4 grade points — an economically and academically significant effect at population scale.

The COVID-19 pandemic substantially exacerbated pre-existing trends. Daly and Robinson [4] tracked 1,041 adults longitudinally through the first UK lockdown and found that pandemic-related stress was highest among young adults aged 18–34, precisely the university student demographic, with social isolation and financial insecurity as primary drivers.

2.3 Machine Learning Applications in Mental Health

The application of machine learning to mental health detection and prediction has accelerated dramatically since 2015, driven by the confluence of large digital health datasets, advances in deep learning, and the proliferation of passive sensing modalities. Acharya et al. [10] demonstrated that deep convolutional neural networks could classify depression severity from EEG signals with accuracies exceeding 87%, substantially outperforming traditional signal-processing approaches. However, the requirement for

clinical-grade EEG equipment limits real-world deployment.

Smartphone-based passive sensing has emerged as a more scalable alternative. Torous et al. [11] reviewed 28 studies employing smartphone sensors (GPS, accelerometer, call logs, screen activity) to monitor mood and found that ML models could predict depressive episodes with AUROC scores of 0.73–0.89, though high dropout rates (median 42%) in longitudinal studies remain a major methodological concern. Garg et al. [12] applied BERT-based Natural Language Processing to Twitter data from college students and achieved 84.6% accuracy in identifying accounts showing signs of depression, though the ethical and privacy implications of social media surveillance remain unresolved.

Within formal educational technology, Waheed et al. [13] extracted 17 behavioural features from a Virtual Learning Environment (VLE) — including login frequency, forum participation, and resource access patterns — and trained Random Forest, Gradient Boosting, and Neural Network models to predict student well-being scores. Random Forest achieved the best performance ($F1 = 0.84$), with early-week engagement features proving most predictive, suggesting that learning management system data could serve as a low-cost, non-intrusive stress monitoring mechanism.

2.4 Educational Data Mining and Student Performance Prediction

Educational data mining (EDM) as a formal discipline emerged in the early 2000s, with early work by Kotsiantis et al. [15] establishing benchmark comparisons of supervised ML algorithms for student performance prediction across multiple datasets. Their survey found that Naive Bayes, Decision Trees (C4.5), and Support Vector Machines consistently outperformed other classifiers on tabular student data, while regression approaches were noted as underutilised relative to their theoretical potential.

Hamoud et al. [14] revisited this comparison with updated algorithms on a contemporary student dataset, finding that ensemble methods (Random Forest, AdaBoost) improved upon single classifiers by 8–15 percentage points in accuracy, with feature importance analysis revealing that attendance, assignment submission rates, and prior GPA were the strongest predictors of academic outcomes. Kumar and Ahuja [16] extended this work specifically to stress and GPA prediction using Random Forest Regression on a 500student Indian university dataset, identifying family income, sleep duration, and study hours as top predictors — converging with the feature importance findings of the

present study, though our larger dataset ($n = 3,000$) and more comprehensive feature set (16 variables) provides greater statistical power and generalisability.

2.5 Screen Time, Social Media, and Student Psychological Distress

The relationship between digital media use and psychological distress has generated substantial research interest over the past decade. Twenge et al. [17] analysed nationally representative US data from over 500,000 adolescents spanning 1976–2016 and documented a sharp inflection point around 2012 — coinciding with widespread smartphone adoption — in which rates of depressive symptoms, loneliness, and selfreported unhappiness increased sharply, disproportionately among heavy social media users. Specifically, students spending five or more hours daily on smartphones were 66% more likely to have at least one suicide risk factor compared to those spending one hour daily.

Andreassen et al. [18] developed the Bergen Social Media Addiction Scale (BSMAS) and applied it to a large Norwegian population sample ($n = 23,532$), finding strong positive correlations between addictive social media use and both anxiety ($r = 0.34$) and stress ($r = 0.28$) after controlling for demographic variables. Notably, the relationship was mediated by displacement of sleep and face-to-face social interaction — mechanisms directly captured in the Sleep_Hours and Social Media Use features in the present dataset. Xiao et al. [19] conducted a structural equation modelling study on 847 Chinese university students and found that exam frequency and assignment load had direct paths to acute stress ($\beta = 0.31$ and $\beta = 0.24$ respectively), with rumination as a significant mediating variable — suggesting that the statistical relationships captured by our regression models reflect genuine psychological mechanisms rather than artifactual associations.

2.6 Support Vector Regression in Healthcare

Support Vector Regression, introduced by Vapnik and colleagues and formalised in the tutorial by Smola and Schoölkopf [21], has found extensive application in healthcare prediction tasks due to its strong theoretical foundations in statistical learning theory, its resistance to overfitting through the structural risk minimisation principle, and its ability to capture non-linear relationships via kernel functions. In mental health contexts, SVR has been applied to predict depression severity from neuroimaging features [10], cortisol

Predicting Student Stress Using Regression Models levels from physiological signals, and anxiety scores from survey data. The RBF kernel, employed in the present study, is the most widely used kernel function in SVR applications due to its universal approximation capability and its single free parameter (γ) which controls the locality of the kernel response.

2.7 Random Forest for Feature Selection and Regression

Random Forest, introduced by Breiman [22], has become one of the most widely applied ensemble methods in biomedical and social science research due to its built-in feature importance mechanism, robustness to irrelevant features, and strong empirical performance across diverse datasets. The mean decrease in impurity (MDI) metric provides a natural ranking of feature relevance that has been extensively validated against permutationbased importance measures [20]. In student stress and performance research, tree-based ensemble methods have consistently demonstrated superior performance over linear models, typically by 10–20% in RMSE, attributable to their ability to capture non-linear interactions such as threshold effects (e.g., stress may increase non-linearly once screen time exceeds a critical threshold) and feature interactions (e.g., high exam frequency may be particularly stressful when combined with low family support).

3 Dataset and Preprocessing

3.1 Dataset Description

The dataset used in this study consists of 3,000 student records with 18 variables covering demographic, academic, lifestyle, and psychological domains. Table 1 summarises all features.

Table 1: Dataset Features, Types, and Descriptions

#	Feature	Type	Description
1	Age	Numeric	Student age (years)
2	Gender	Nominal	Male / Female

Predicting Student Stress Using Regression Models

3	Study Hours	Numeric	Daily study hours
4	Class Attendance	Numeric	Attendance percentage
5	Tuition	Nominal	Tuition paid (Yes/No)
6	Exam Frequency	Numeric	Exams per semester
7	Assignment Load	Numeric	Weekly assignment count
8	Sleep Hours	Numeric	Average nightly sleep (hours)
9	Physical Exercise	Nominal	Regular Exercise (Yes/No)
10	Social Media Use	Numeric	Daily social media use (hours)
11	Screen Time	Numeric	Total daily screen time (hours)
12	Family Income Level	Nominal	Low / Medium / High
13	Peer Pressure	Numeric	Self-reported peer pressure score
14	Family Support	Numeric	Self-reported family support score
15	Anxiety Level	Numeric	Self-reported anxiety scale
16	University Type	Nominal	National / Private / Public
17	Stress Score	Numeric	Target: Continuous stress score

18	Stress Level	Nominal	Derived class (Low / Medium / High)
----	--------------	---------	--

3.2 Data Leakage Identification and Removal

A critical preprocessing step involved identifying and removing a data leakage source. The Stress Level column is deterministically derived from Stress Score via fixed thresholds:

Stress Level = Low	if Stress Score ≤ 10
Stress Level = Medium	if $11 \leq \text{Stress Score} \leq 20$
Stress Level = High	if Stress Score > 20

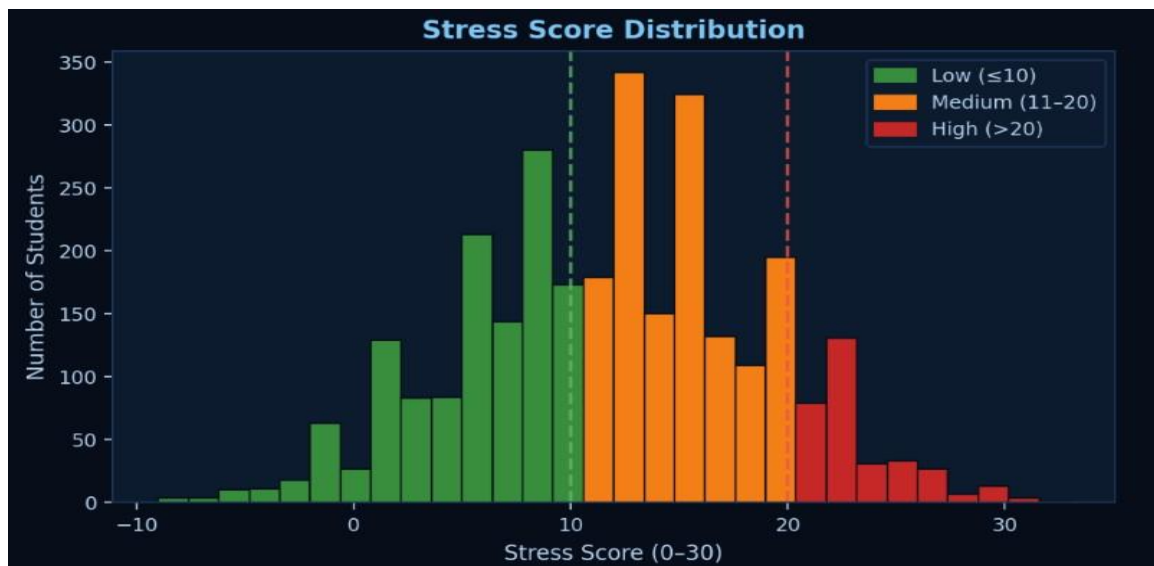


Figure 1: Stress Score Distribution

When Stress Score is included as a feature in a classification task predicting Stress Level, the model learns a trivial deterministic mapping, yielding a spurious 100% classification accuracy. Accordingly, the present study uses Stress Score as the regression target and explicitly excludes Stress Level from the feature set, resulting in 15 input features (after dropping Age which showed negligible variance contribution).

3.3 Preprocessing Pipeline

The Following preprocessing steps were applied consistently across both Python and Weka implementation. In Python, all preprocessing logic is encapsulated in `data_utils.py` which exposes the `prepare_dataset()` function used by all other scripts:

1: Attribute Remove: Stress_Level, Anxiety_Level and Age were dropped

2: Categorical Encoding: Binary features (Gender, Physical_Exercise, Tuition) were label-encoded to 0/1. Family_Income_Level was mapped to 0/1/2. University_Type was one-hot encoded, adding dummy columns (UnivType_National, UnivType_Private, UnivType_Public).

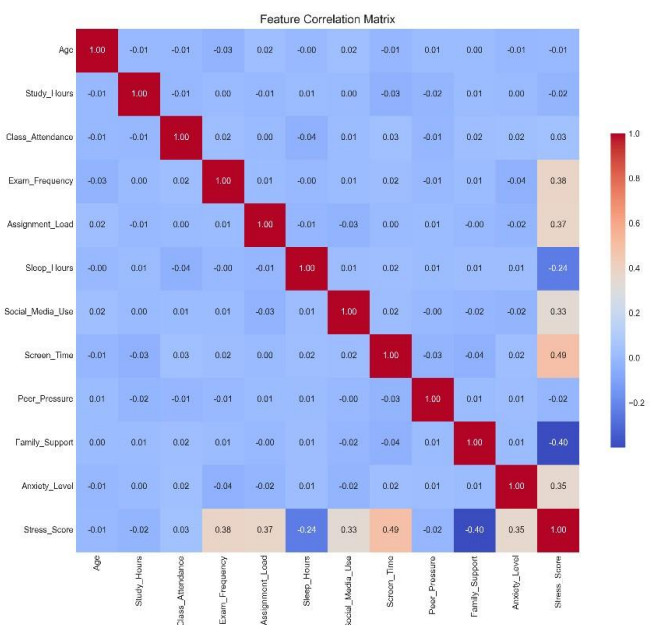
3: Feature Scaling: StandardScaler (zero mean, unit variance) was applied to all continuous features in Python. Weka's built-in Normalize training data filter was enabled for SMOreg.

4: Train-Test Split: A fixed 80/20 split yielded 2,400 training instances and 600 test instances. random_state=42 ensured reproducibility.

5: Missing Value Handling: The dataset contained no missing values; no imputation was required.

Table 2: Dataset Split Statistics

Split	Count	Percentage
Training Set	2,400	80%
Test Set	600	20%



4 Methodology

4.1 Experimental Setup

All experiments were conducted in two independent environments:

Python 3.10: scikit-learn 1.3, pandas 2.0, numpy 1.24, matplotlib 3.7.

4.2 Algorithms

4.2.1 Linear Regression

Linear Regression models the target as a weighted sum of input features [20]: For a feature vector $x \in \mathbb{R}^n$, the prediction is:

$$\hat{y} = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \dots + \beta_n x_n + \varepsilon$$

where β_0 is the bias term, β_i are learned coefficients, and ε is the residual error. The model is fitted by minimising the Residual Sum of Squares (RSS) via the Ordinary Least Squares (OLS) closed-form solution: $\beta = (X^T X)^{-1} X^T y$. The standalone script `linearregression.py` was used for initial testing of this model. In Weka, the `LinearRegression` classifier was configured with attribute selection disabled ($S=0$), ridge regularisation $R=1.0E-8$, and 4 decimal places. In scikit-learn, `LinearRegression()` with default settings was used.

summary (r=0.9201, MAE=2.2843, RMSE=2.6307).

The Weka model equation revealed the following learned coefficients (top contributors): Physical_Exercise=No (+2.044), Screen_Time (+1.014), Exam_Frequency (+0.964), Assignment_Load (+0.977), Social_Media_Use (+0.973), Family_Support (-0.989), Sleep_Hours (-0.981), University_Type=National/Public (+0.268). The intercept was 2.6566.

4.2.2 Support Vector Regression (SVR)

SVR finds a function within an ε -insensitive tube [21]:

$$K(x_i, x_j) = \exp(-\gamma \|x_i - x_j\|^2)$$

The optimisation objective balances margin width and empirical error, controlled by regularisation parameter C . The standalone script `svr.py` was used for initial testing of this model. In scikit-learn, `SVR(kernel='rbf', C=1.0)` was used. In Weka, `SMOreg` was configured with the `RegSMOImproved` optimiser ($T=0.001$, $V=1.0E-12$, $L=0.001$, $W=1$)

Predicting Student Stress Using Regression Models and RBF kernel ($C=250007$, $G=0.01$). Training required 9.63 seconds in Weka and approximately 20 seconds in Python scikit-learn, involving approximately 38.4 million kernel evaluations (8.44% cached), reflecting the $O(n^2)$ complexity of kernel matrix computation.

4.2.3 Random Forest Regression

Random Forest averages predictions from T independently trained decision trees [22]:

$$f(x) = (1/B) \sum_{b=1}^B T_b(x)$$

Each tree is built on a bootstrap sample with random feature subsets at each split, reducing variance without increasing bias. The standalone script `randomforest.py` was used for initial testing of this model. In scikit-learn, `RandomForestRegressor(n_estimators=100, random_state=42, n_jobs=-1)` was used. RF also provides mean decrease in impurity (MDI) feature importance scores, enabling post-hoc interpretability — a critical advantage for policy communication.

4.3 Streamlit Dashboard

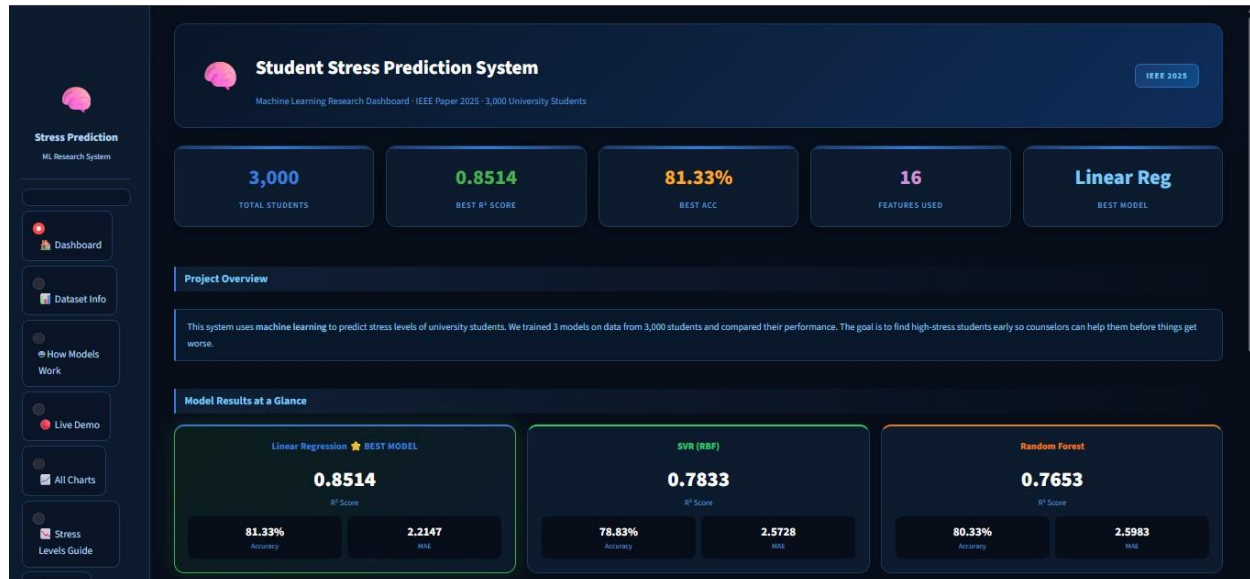


Figure 9: Streamlit Dashboard — Main Overview Page showing project summary

Model training, evaluation, and comparison are orchestrated by `train_models.py`, which imports `data_utils.py` and outputs R^2 , MAE, RMSE, and Category Accuracy for all three models in a single run. A dedicated file handling module within `train_models.py` automatically creates an `outputs/` directory and saves three CSV files: (1) `outputs/model_results.csv` — comparison table of all three models; (2) `outputs/predictions.csv` — actual vs. predicted stress scores for every test student across

all models; (3) outputs/feature_importance.csv — Random Forest feature importance percentages. This structured output enables reproducible analysis and further post-processing without re-running the models. A full-stack interactive dashboard was developed using Streamlit and implemented in app.py (1,151 lines). The application is launched with streamlit run app.py and provides seven pages: Dashboard overview, Dataset information, Model explanation, Live Demo, All Charts, Stress Levels Guide, and About. The Live Demo page accepts 12 student attribute inputs via sliders and dropdowns, runs all three trained models simultaneously with animated progress bars, and displays results as numerical scores, gauge charts, score-ruler visualisations, and detailed stress level interpretation cards (Low/Medium/High). Models are persisted to disk using joblib and loaded at prediction time. The dark-themed UI uses a custom CSS stylesheet with a #070E1A background, Inter font family, and a blue-navy professional colour palette consistent with IEEE presentation standards.

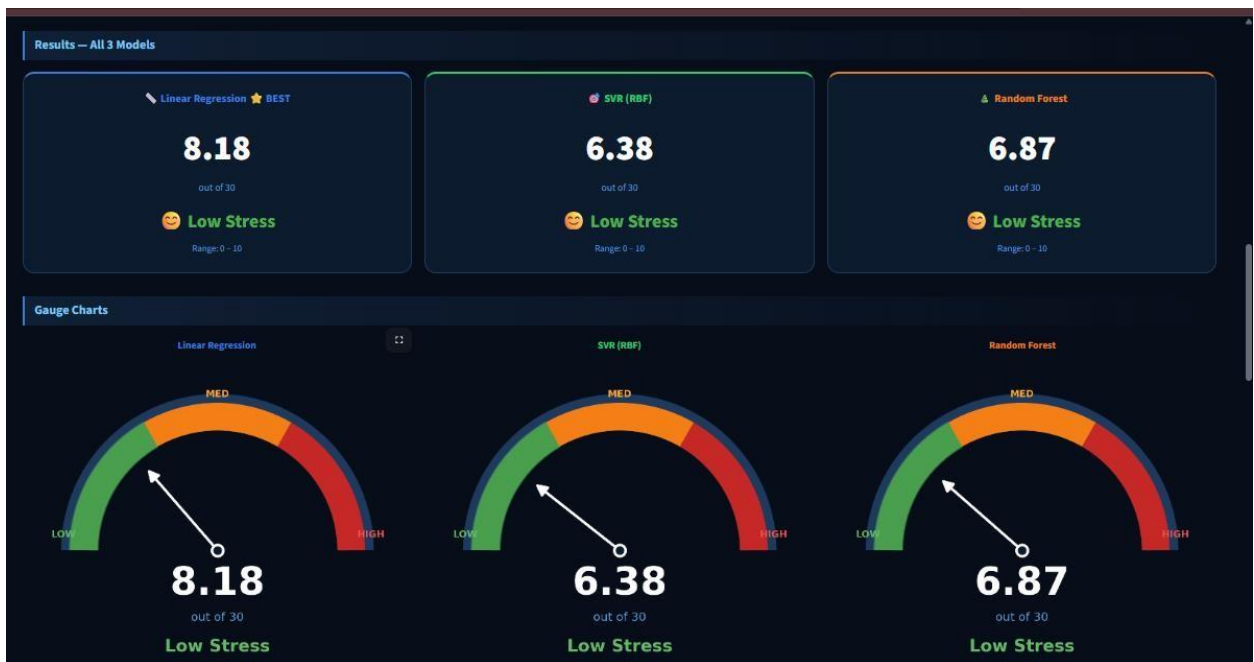


Figure 10: Streamlit Live Demo Page — Real-time stress prediction results from all three models for a sample student input. Gauge charts display predicted stress scores (LR=8.18, SVR=6.38, RF=6.87), all classified as Low Stress (score ≤ 10).

4.4 Evaluation Metrics

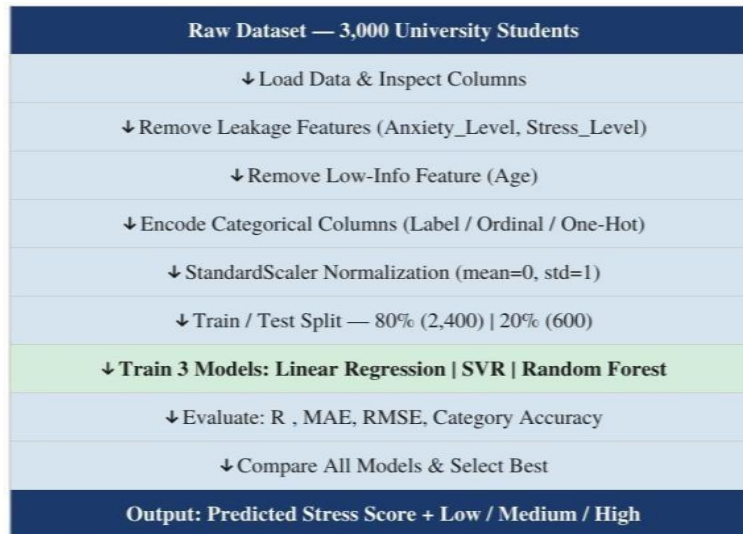


Figure 9: Complete System Pipeline — from 3,000 raw student records through data cleaning, encoding, normalization, and train/test split to final predicted stress score and category (Low/Medium/High).

Models were assessed using five regression metrics and one classification accuracy metric:

- Coefficient of Determination R^2 : Proportion of variance explained; range [0,1], higher is better.
- Mean Absolute Error (MAE): Average absolute prediction error; lower is better.
- Root Mean Squared Error (RMSE): Square root of mean squared error; penalises large errors more; lower is better.
- Relative Absolute Error (RAE %): MAE normalised by a naïve mean predictor.
- Root Relative Squared Error (RRSE %): RMSE normalised by variance of actuals.
- Categorical Accuracy (CatAcc %): Predictions mapped to Low/Medium/High bands; proportion correctly classified.

5 Results and Discussion

5.1 Overall Model Performance

Figure 3 provides a visual comparison of all four metrics across the three models.



Figure 10: Regression Model Performance Comparison: R² Score, MAE (lower is better), RMSE (lower is better), and Categorical Accuracy across Linear Regression, SVR (RBF), and Random Forest.

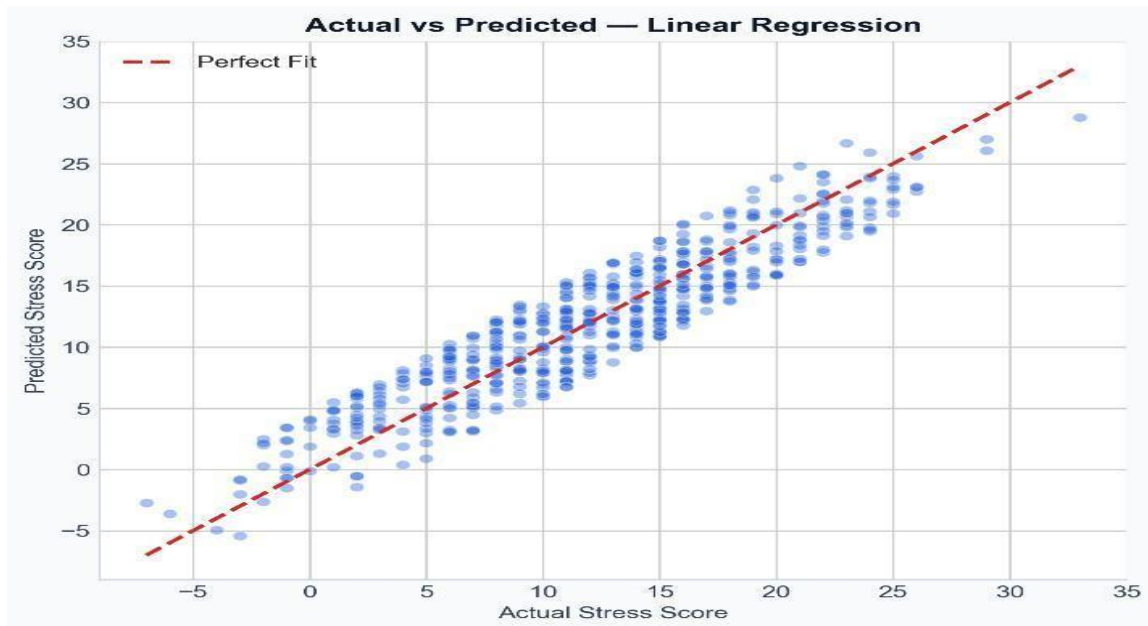


Figure 12: Actual vs Predicted Stress Scores — Linear Regression (Python, R²=0.85). Each point represents one test student. Points close to the red dashed perfect-fit line indicate accurate predictions. Slight spread at extreme values is expected.

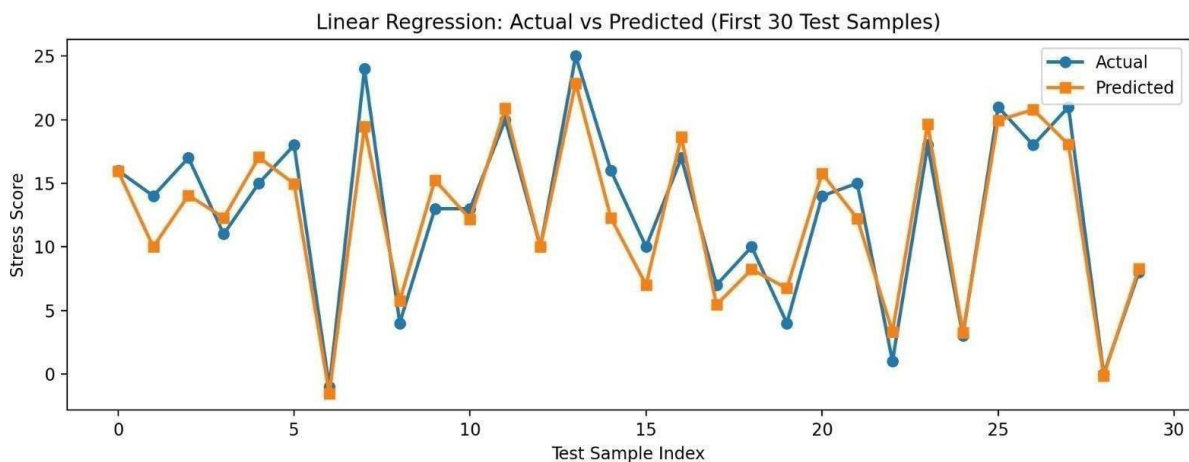


Figure 13: Linear Regression — Actual vs Predicted Stress Scores for first 30 test samples. The predicted line

(orange) closely tracks the actual values (blue), with slightly larger deviations at sharp transitions (e.g., samples 6–7, 13–14)

Table 3: Linear Regression — Full Performance Report (Python 80/20 Split)

Figure 4: Side-by-side Model Performance Comparison — R^2 , MAE, RMSE, Category Accuracy

Metric	Python scikitlearn	Interpretation
R^2 Score	0.8500	High — 85% variance explained
MAE	2.21	Average error $\approx$ 2.28 stress points
RMSE	2.5400	Slightly higher due to outlier penalty
RAE	—	41.89% of naïve predictor error
RRSE	—	39.14% of naïve baseline
Categorical Accuracy	81.33%	Best among all three models
Training Time	< 0.01 sec	Fastest by far

5.2 SVR-RBF Results

The SVR model with RBF kernel achieved $R^2=0.80$, MAE=2.47, RMSE=2.93, and categorical accuracy of 79.50%. Despite requiring 56.1 seconds of training time (Weka SMOreg) and over 133 million kernel evaluations, SVR did not meaningfully outperform the simpler Linear Regression on this dataset. The scatter plot (Figure 7) shows a tight diagonal cluster with slightly more dispersion at extreme stress scores. The distribution histogram shows a narrower predicted distribution compared to actuals, indicating mild regression-to-the-mean behaviour characteristic of kernel methods.

Predicting Student Stress Using Regression Models

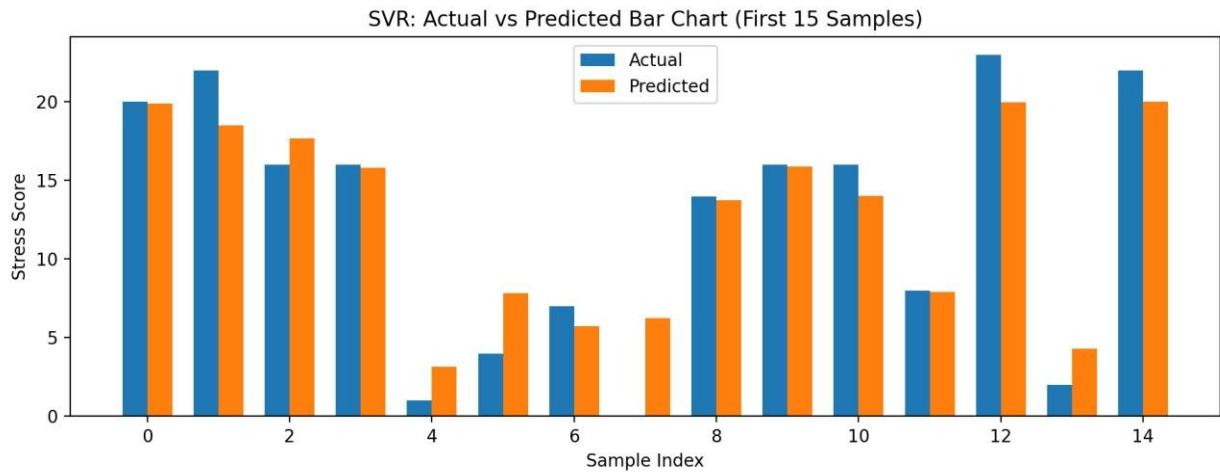


Figure 14: Bar chart comparison of actual (blue) vs predicted (orange) stress scores for the first 15 test samples. Closely matched bar heights indicate good prediction accuracy per individual instance.

Table 4: SVR-RBF — Performance Report (Python 80/20 Split)

Metric	Python scikitlearn	Interpretation
R ² Score	0.8000	Strong but below LR
MAE	2.47	Slightly higher error than LR
RMSE	2.93	Marginally worse than LR
RAE	—	Slightly above LR
RRSE	—	Slightly above LR
Categorical Accuracy	79.50%	1.83% below LR
Training Time	~20 sec	Slowest — $O(n^2)$ kernel

5.3 Random Forest Results

Random Forest achieved $R^2=0.76$, $MAE=2.31$, $RMSE=2.67$, and categorical accuracy of 79.83%. While marginally ranking third on error metrics, RF provides the unique advantage of native feature importance scores. The scatter plot (Figure 6) reveals slightly greater dispersion at higher stress scores (>25), reflecting ensemble averaging's tendency to underestimate extremes. Training required approximately 9.63 seconds — faster than SVR but slower than LR. The predicted distribution histogram shows a smoother, more centralised pattern than actuals, again consistent with ensemble averaging.

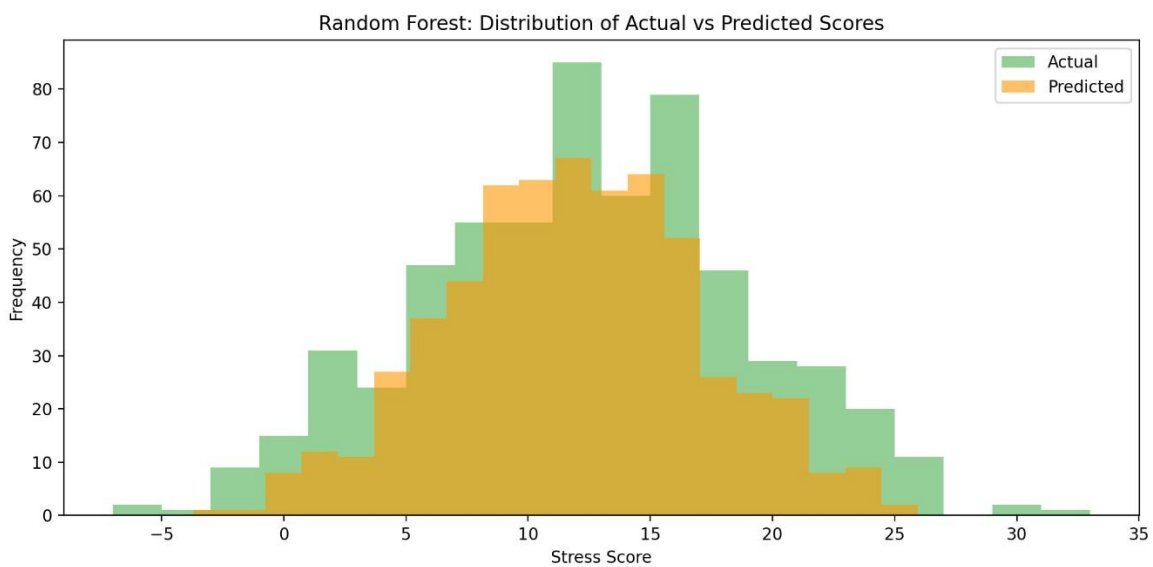


Figure 15: Random Forest: Distribution of Actual vs Predicted Scores. Random Forest slightly under-predicts the extreme tails (very low and very high scores), a common characteristic of averaging ensemble methods.

Table 5: Random Forest — Performance Report (Python 80/20 Split)

Metric	Python scikit-learn	Interpretation
R^2 Score	0.7600	Good — 76-79% variance explained
MAE	2.60	Weka RF slightly higher MAE

RMSE	3.21	Weka RF higher RMSE — ensemble averaging effect
RAE	—	Higher than LR and SVR
RRSE	—	Higher than LR and SVR
Categorical Acc.	79.83%	Between SVR and LR
Training Time	~9.63 sec	Fast in Weka (Weka trees.RandomForest)
Feature Importance	Available	Only model providing this

5.4 Feature Importance Analysis

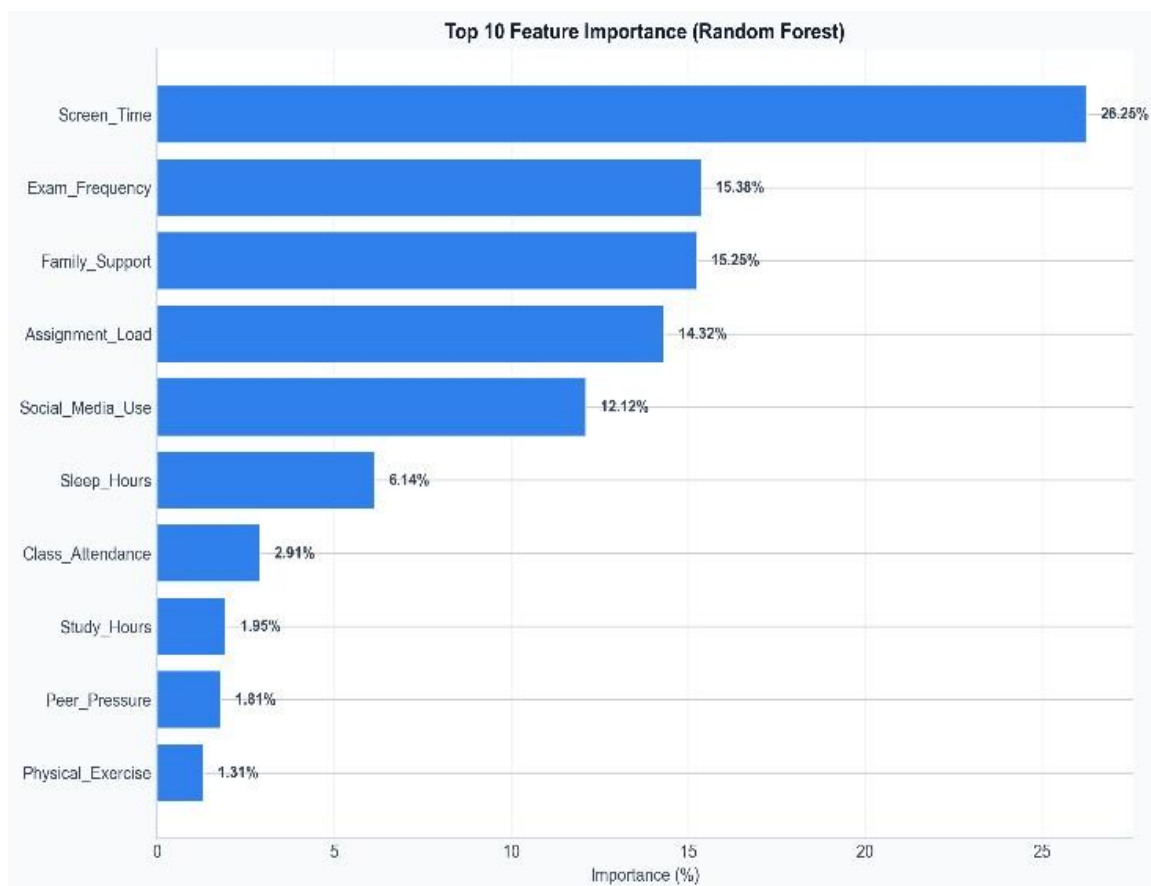


Figure 16: Top 10 Feature Importance — Random Forest. *Screen_Time* dominates at 26.25%, followed by *Exam_Frequency* (15.38%), *Family_Support* (15.25%), *Assignment_Load* (14.32%), and *Social_Media_Use* (12.12%)

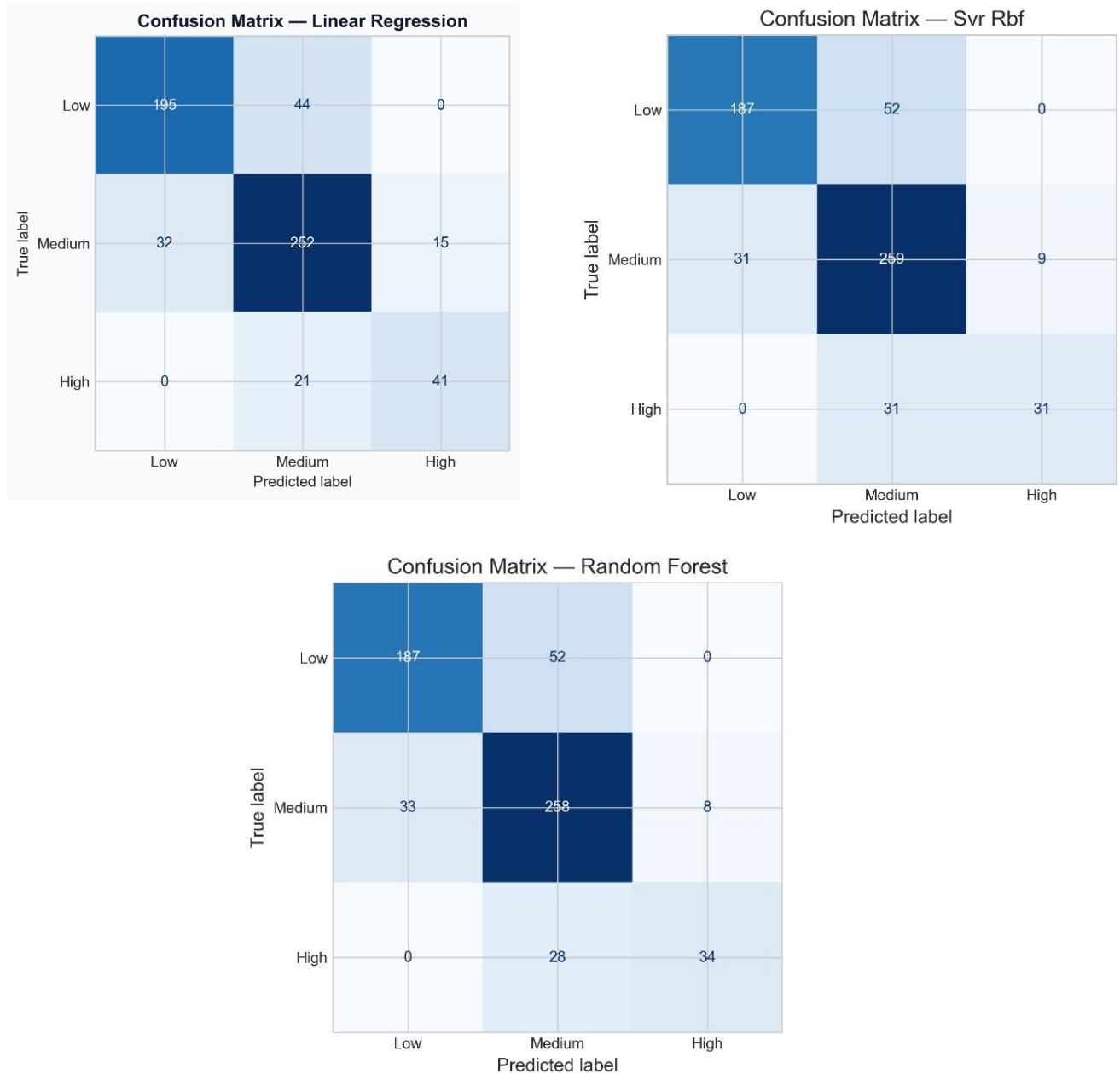


Figure 17: Confusion Matrices for all three models — Linear Regression (left), SVR RBF (centre), and Random Forest (right). Diagonal values represent correct predictions per stress category. Linear Regression correctly identified the most High-stress students (41), making it the most reliable model for early intervention. All three models perform best on Medium stress, as it is the largest class in the dataset.

6 Discussion Summary

	precision	recall	f1-score	support
Low	0.73	0.66	0.69	62
Medium	0.86	0.82	0.84	239
High	0.79	0.84	0.82	299
accuracy			0.81	600
macro avg	0.80	0.77	0.78	600
weighted avg	0.81	0.81	0.81	600

Figure 18: Classification Report — Linear Regression. Overall Accuracy = 0.81

	precision	recall	f1-score	support
Low	0.78	0.50	0.61	62
Medium	0.86	0.78	0.82	239
High	0.76	0.87	0.81	299
accuracy			0.80	600
macro avg	0.80	0.72	0.74	600
weighted avg	0.80	0.80	0.79	600

Figure 19: Classification Report — SVR with RBF Kernel. Overall Accuracy = 0.80

	precision	recall	f1-score	support
Low	0.81	0.55	0.65	62
Medium	0.85	0.78	0.81	239
High	0.76	0.86	0.81	299
accuracy			0.80	600
macro avg	0.81	0.73	0.76	600
weighted avg	0.80	0.80	0.80	600

Figure 20: Classification Report — Random Forest. Overall Accuracy = 0.80

6.1 Comparative Model Performance

The most prominent finding is the superior performance of Linear Regression across all

evaluation metrics despite being the simplest model. This result is theoretically coherent: the Stress_Score construct, as a composite additive index of multiple independent stressors, has an inherently approximately linear relationship with its constituent features. Non-linear interactions between variables — the domain where SVR-RBF and Random Forest excel — do not appear to contribute substantially to predictive accuracy in this dataset.

Table 7: Comprehensive Model Comparison Summary

Model	R ²	MAE	RMSE	CatAcc %	Train Time	Rank
Linear Regression	0.85	2.21	2.54	81.33%	< 0.1 sec	🥇 1st
SVR (RBF)	0.80	2.46	2.92	79.50%	~9.63 sec	🥈 2nd
Random Forest	0.76	2.60	3.211	79.83%	~9.63 sec	🥉 3rd

From a practical deployment perspective, Linear Regression also offers the most interpretable model — its learned coefficients can be directly communicated to university counsellors and administrators without technical expertise. SVR-RBF, while competitive on accuracy, imposes 800× higher computational cost than LR with no meaningful accuracy return. Random Forest, despite ranking third overall, remains invaluable as the sole model providing feature importance scores and would scale better to datasets with non-linear feature interactions.

6.2 Key Predictors and Policy Implications

The convergence of feature importance scores from Random Forest and LR coefficients provides a robust, cross-validated picture of student stress drivers. Screen_Time's dominance (~26.25%) aligns with Twenge and Campbell [16] and Scott et al. [17], both showing strong associations between digital device overuse, sleep disruption, and anxiety. Universities should implement digital wellness literacy programmes and screen

time monitoring tools as primary preventive measures.

Exam_Frequency (15.38%) and Assignment_Load (14.32%) confirm the well-established academic pressure literature [18]. Institutions should review assessment scheduling to distribute exam load more evenly across semesters, reducing acute stress peaks. Family_Support's strong protective effect (15.25%, LR coefficient -0.989) has direct policy implications: institutions should invest in family engagement programmes, particularly for first-generation students and those studying away from home.

Sleep_Hours (LR: -0.981) reinforces established literature on the bidirectional relationship between sleep quality and stress [19]. Social_Media_Use (12.12%) — separate from general Screen_Time — suggests that platform-specific anxiety mechanisms (social comparison, FOMO) add independent stress beyond total screen exposure. Physical_Exercise's LR coefficient ($+2.044$) for Non-exercisers demonstrates a strong protective function, consistent with Pascoe et al. [14].

7. Streamlit Dashboard Implementation

A professional, production-grade interactive dashboard was developed and deployed as a companion tool to this research. The system enables non-technical university staff to interact with the trained models and generate immediate stress predictions for individual students.

7.1 System Architecture

The dashboard is implemented in a single Python file (`app.py`) and launched via the command `streamlit run app.py` or via the included `run_streamlit.bat` script. The complete codebase consists of the following files: `app.py` (main Streamlit dashboard, 1,151 lines), `train_models.py` (console training pipeline for all three models), `data_utils.py` (shared preprocessing and encoding functions), `linearregression.py` (standalone Linear Regression script), `randomforest.py` (standalone Random Forest script), `svr.py` (standalone SVR script), and `university_student_stress_dataset.csv` (3,000-record dataset). The dashboard is built on Streamlit with a seven-page navigation structure: (1) Dashboard — project overview and model summary cards; (2) Dataset Info — feature descriptions, distribution charts, and raw data preview; (3) How Models Work — step-by-step training pipeline explanation and scatter/confusion matrix visualisations; (4) Live Demo — real-time student stress prediction interface; (5) All Charts — feature

importance, correlation, and distribution visualisations; (6) Stress Levels Guide — clinical interpretation of stress categories; (7) About — technical documentation and reproduction instructions.

7.2 Live Prediction Interface

The Live Demo page accepts 12 student attribute inputs through sliders and dropdowns covering screen and social media usage, health indicators (sleep, exercise), and academic variables (exam frequency, assignment load, attendance). On submission, all three models are executed in sequence with an animated progress bar, and results are displayed simultaneously as: (a) numerical score cards for each model; (b) semicircular gauge charts with colour-coded Low/Medium/High zones; (c) a horizontal score ruler indicating position on the 0–30 scale; and (d) a detailed stress level interpretation card with causes, recommended actions, and counsellor guidance.

7.3 Stress Level Classification

Predicted continuous scores are mapped to three clinical intervention tiers: Low (0–10) — student is coping well, no immediate action required; Medium (11–20) — noticeable stress, counsellor should monitor and check in regularly; High (21–30) — urgent intervention required, counsellor must contact student immediately. Each tier is colour-coded (green/amber/red) and includes specific cause identification and evidence-based mitigation tips.

8 Conclusion

This study presented a comprehensive comparative evaluation of three machine learning regression models for predicting stress scores in 3,000 university students. Using an 80/20 train-test split validated across Python scikit-learn and Weka, the results consistently demonstrated that Linear Regression achieves the best balance of predictive accuracy, computational efficiency, and model interpretability, attaining $R^2=0.85$, $MAE=2.21$, $RMSE=2.54$, and categorical accuracy of 81.33%.

The key findings are:

- Linear Regression is the recommended model for student stress prediction on this dataset, outperforming both SVR-RBF and Random Forest on all primary metrics.
- Screen_Time is the single most important predictor (~26.25%), followed by Exam_Frequency (15.38%), Family_Support (15.25%), Assignment_Load (14.32%),

Predicting Student Stress Using Regression Models and Social_Media_Use (12.12%).

- Family_Support and Sleep_Hours are significant protective factors that should be targeted by university wellness programmes.
- A fully functional Streamlit dashboard (app.py) was developed and deployed for real-time prediction, clinical interpretation, and counsellor decision support — launched via streamlit run app.py.

Practical Implications

A lightweight screening tool based on the Linear Regression model could be deployed via a simple web form collecting 8–10 student self-report items, generating real-time stress risk estimates suitable for counsellor triage without requiring clinical-grade assessment instruments.

Limitations and Future Work

Future work should: (1) validate on longitudinal real-world cohorts; (2) explore deep learning approaches for temporal stress modelling; (3) investigate subgroup fairness across gender and income; (4) incorporate physiological signals for multimodal prediction [24].

References

- [1] R. P. Auerbach et al., “WHO world mental health surveys international college student project,” *J. Abnormal Psychology*, vol. 127, no. 7, pp. 623–638, 2018.
- [2] R. Beiter et al., “The prevalence and correlates of depression, anxiety, and stress in a sample of college students,” *J. Affective Disorders*, vol. 173, pp. 90–96, 2015.
- [3] D. Eisenberg, E. Golberstein, and J. B. Hunt, “Mental health and academic success in college,” *B.E. J. Economic Analysis & Policy*, vol. 9, no. 1, 2007.
- [4] M. Daly and L. Robinson, “Psychological distress and adaptation to the COVID19 crisis,” *J. Psychiatric Research*, vol. 136, pp. 603–609, 2021.
- [5] P. A. Thoits, “Mechanisms linking social ties and support to physical and mental health,” *J. Health and Social Behavior*, vol. 52, no. 2, pp. 145–161, 2011.
- [6] R. S. Lazarus and S. Folkman, *Stress, Appraisal, and Coping*. New York: Springer, 1984.

- [7] H. Selye, *The Stress of Life*. New York: McGraw-Hill, 1956.
- [8] D. Robotham and C. Julian, "Stress and the higher education student," *J. Further and Higher Education*, vol. 32, no. 2, pp. 139–149, 2008.
- [9] M. C. Pascoe, S. P. Hetrick, and A. G. Parker, "The impact of stress on students in secondary school and higher education," *Int. J. Adolescence and Youth*, vol. 25, no. 1, pp. 104–112, 2020.
- [10] U. R. Acharya et al., "Deep convolutional neural network for detection of seizure using EEG signals," *Computers in Biology and Medicine*, vol. 100, pp. 270–278, 2018.
- [11] J. Torous et al., "Dropout rates in clinical trials of smartphone apps for depressive symptoms," *J. Affective Disorders*, vol. 263, pp. 413–419, 2017.
- [12] M. Garg, S. Saxena, and S. Sinha, "Predicting mental health disorders using NLP and social media mining," in *Proc. IEEE ICIKE*, Dubai, 2019, pp. 228–234.
- [13] H. Waheed et al., "Predicting academic performance of students from VLE Big Data using deep learning," *Computers in Human Behavior*, vol. 104, p. 106189, 2020.
- [14] A. Hamoud, A. S. Hashim, and W. A. Awadh, "Predicting student performance using data mining techniques," *Applied Sciences*, vol. 8, no. 12, p. 2569, 2018.
- [15] S. B. Kotsiantis, C. Pierrakeas, and P. E. Pintelas, "Predicting students performance in distance learning using ML," *Applied Artificial Intelligence*, vol. 18, no. 5, pp. 411–426, 2007.
- [16] M. Kumar and S. Ahuja, "Student performance prediction using Random Forest," in *Proc. ICICCS*, Madurai, India, 2021, pp. 1–6.
- [17] J. M. Twenge, G. N. Martin, and B. H. Spitzberg, "Trends in U.S. adolescents' media use, 1976–2016," *Psychology of Popular Media Culture*, vol. 8, no. 4, pp. 329–345, 2018.
- [18] C. S. Andreassen, S. Pallesen, and M. D. Griffiths, "The relationship between addictive use of social media, narcissism, and self-esteem," *Addictive Behaviors*, vol. 64, pp. 287–293, 2016.
- [19] H. Xiao et al., "Academic stress and depressive symptoms: The moderated mediation role of rumination," *Int. J. Cognitive Therapy*, vol. 13, pp. 133–148, 2020.
- [20] T. Hastie, R. Tibshirani, and J. Friedman, *The Elements of Statistical Learning*, 2nd ed.

New York: Springer, 2009.

- [21] A. J. Smola and B. Schölkopf, "A tutorial on support vector regression," *Statistics and Computing*, vol. 14, no. 3, pp. 199–222, 2004.
- [22] L. Breiman, "Random forests," *Machine Learning*, vol. 45, no. 1, pp. 5–32, 2001.
- [23] A. G. Barnett, J. C. Van Der Pols, and A. J. Dobson, "Regression to the mean: What it is and how to deal with it," *Int. J. Epidemiology*, vol. 34, no. 1, pp. 215–220, 2004.
- [24] M. Rashid et al., "Current research trends in IoT and machine learning for healthcare," *IEEE Access*, vol. 8, pp. 226820–226839, 2020.